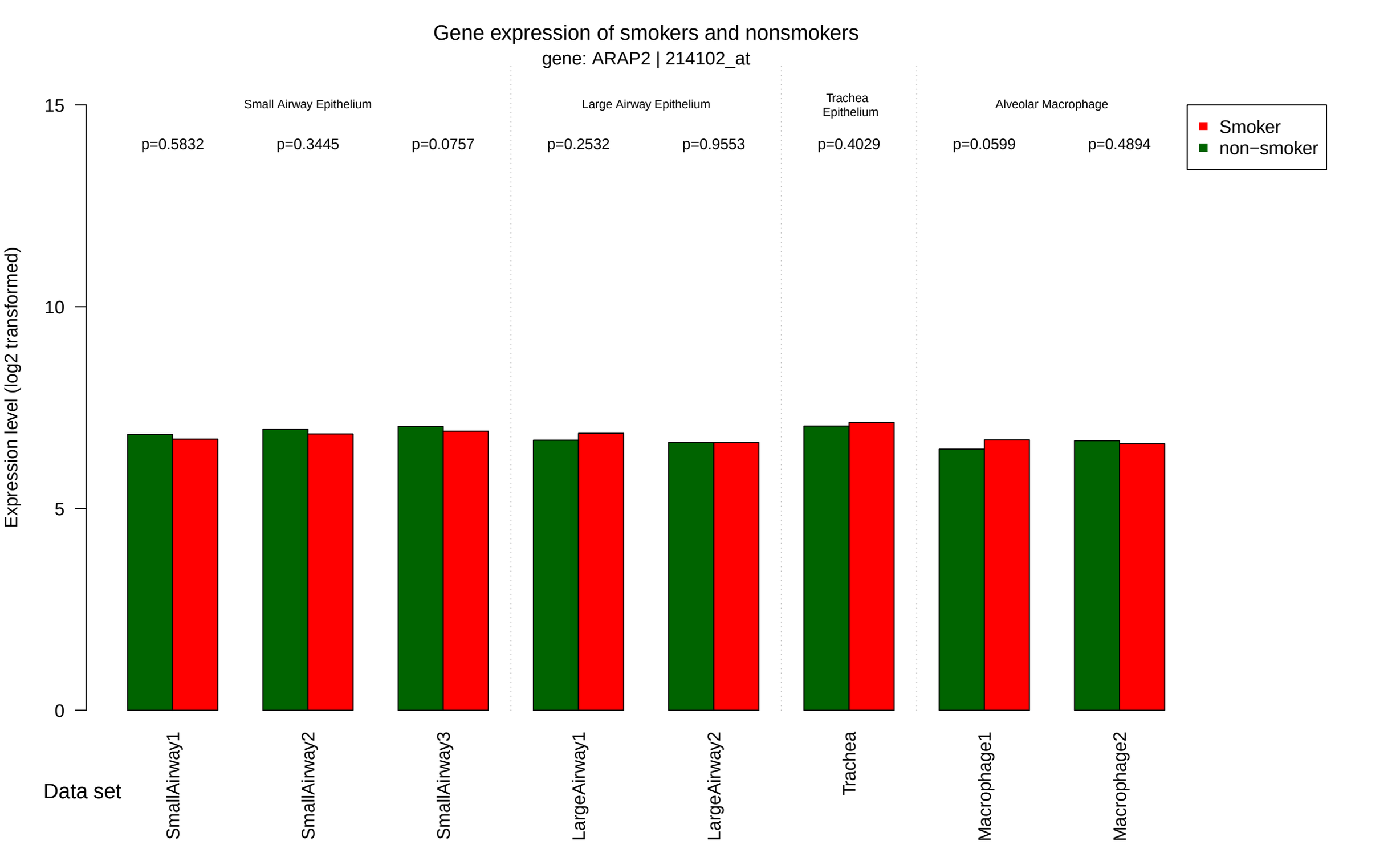
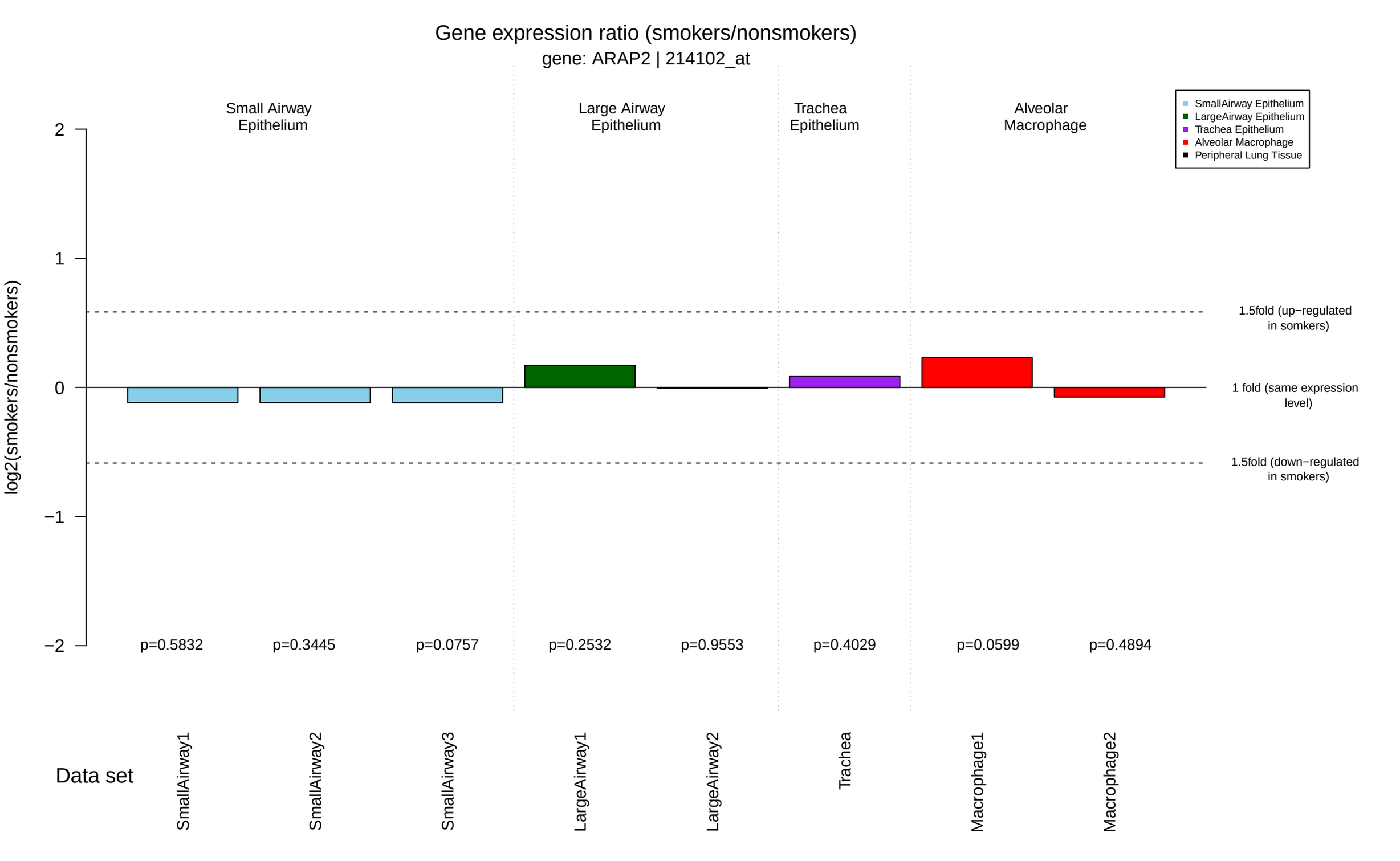
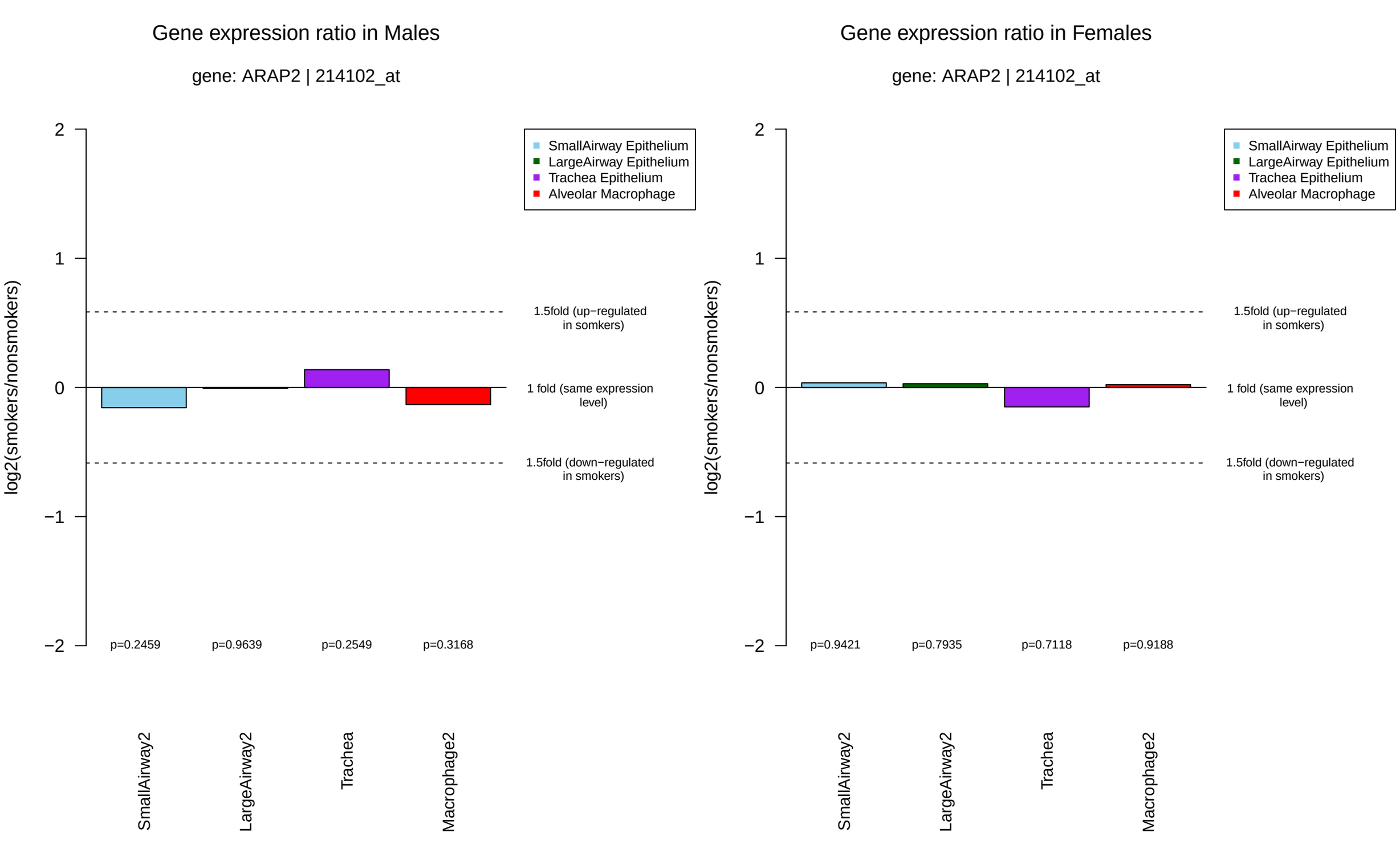
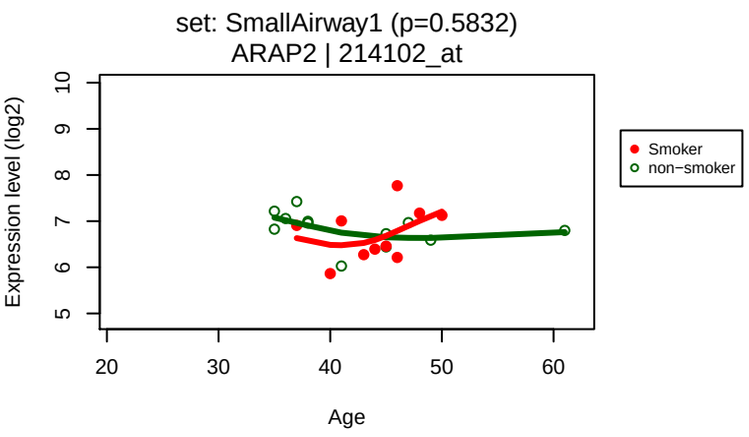
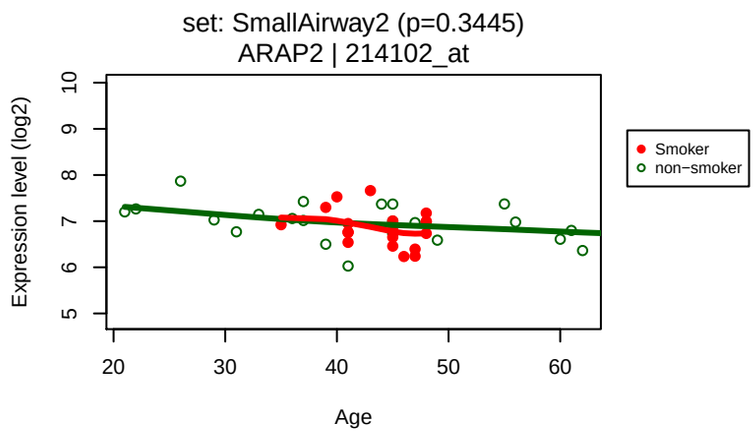
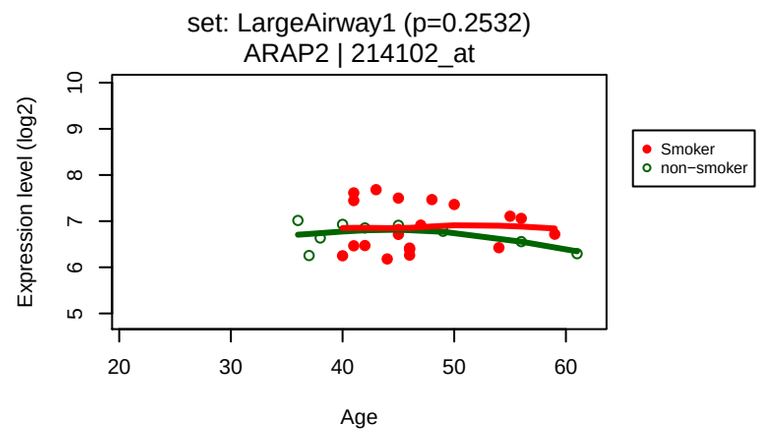

Gene expression of smokers and nonsmokers
gene: ARAP2 | 214102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5832
p=0.3445
p=0.0757
p=0.2532
p=0.9553
p=0.4029
p=0.0599
p=0.4894
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

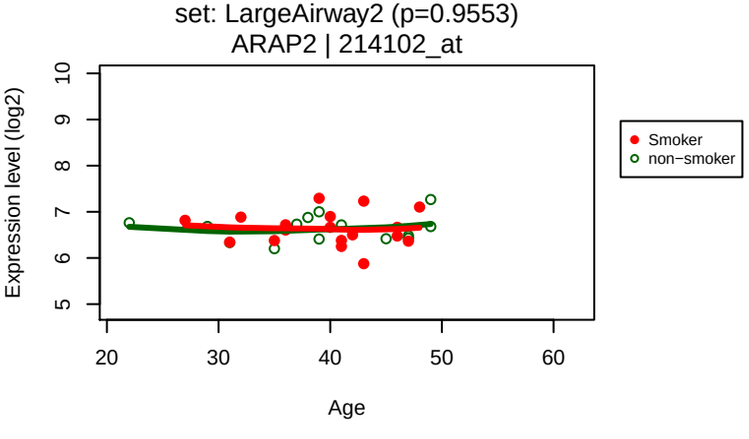

Gene expression ratio (smokers/nonsmokers)
gene: ARAP2 | 214102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5832
p=0.3445
p=0.0757
p=0.2532
p=0.9553
p=0.4029
p=0.0599
p=0.4894
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

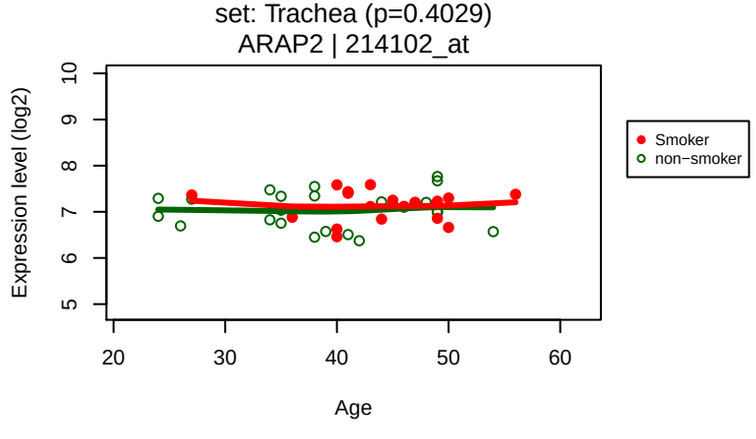

Gene expression ratio in Males
gene: ARAP2 | 214102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2459
p=0.9639
p=0.2549
p=0.3168
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARAP2 | 214102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9421
p=0.7935
p=0.7118
p=0.9188
SmallAirway2
LargeAirway2
Trachea
Macrophage2

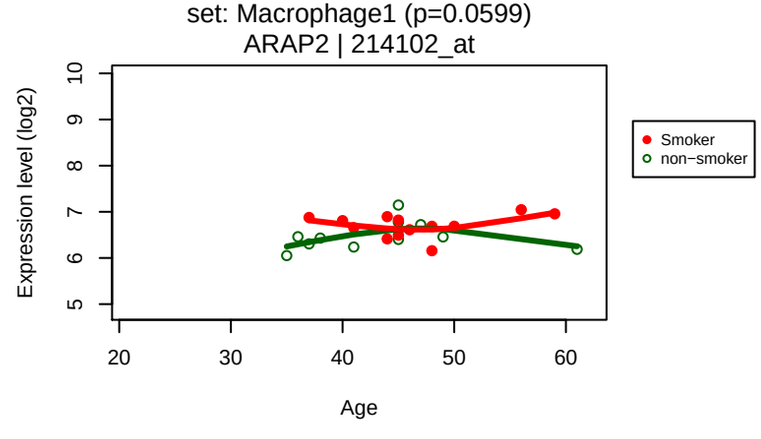

set: SmallAirway1 (p=0.5832)
ARAP2 | 214102_at
Expression level (log2)
Age
Smoker
non−smoker

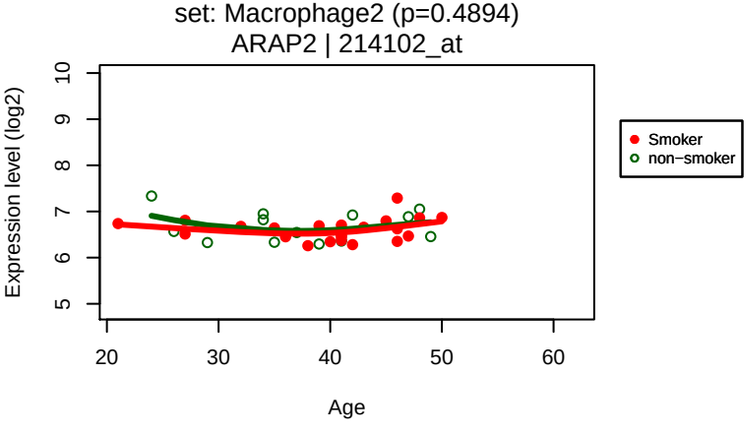

set: SmallAirway2 (p=0.3445)
ARAP2 | 214102_at
Expression level (log2)
Age
Smoker
non−smoker

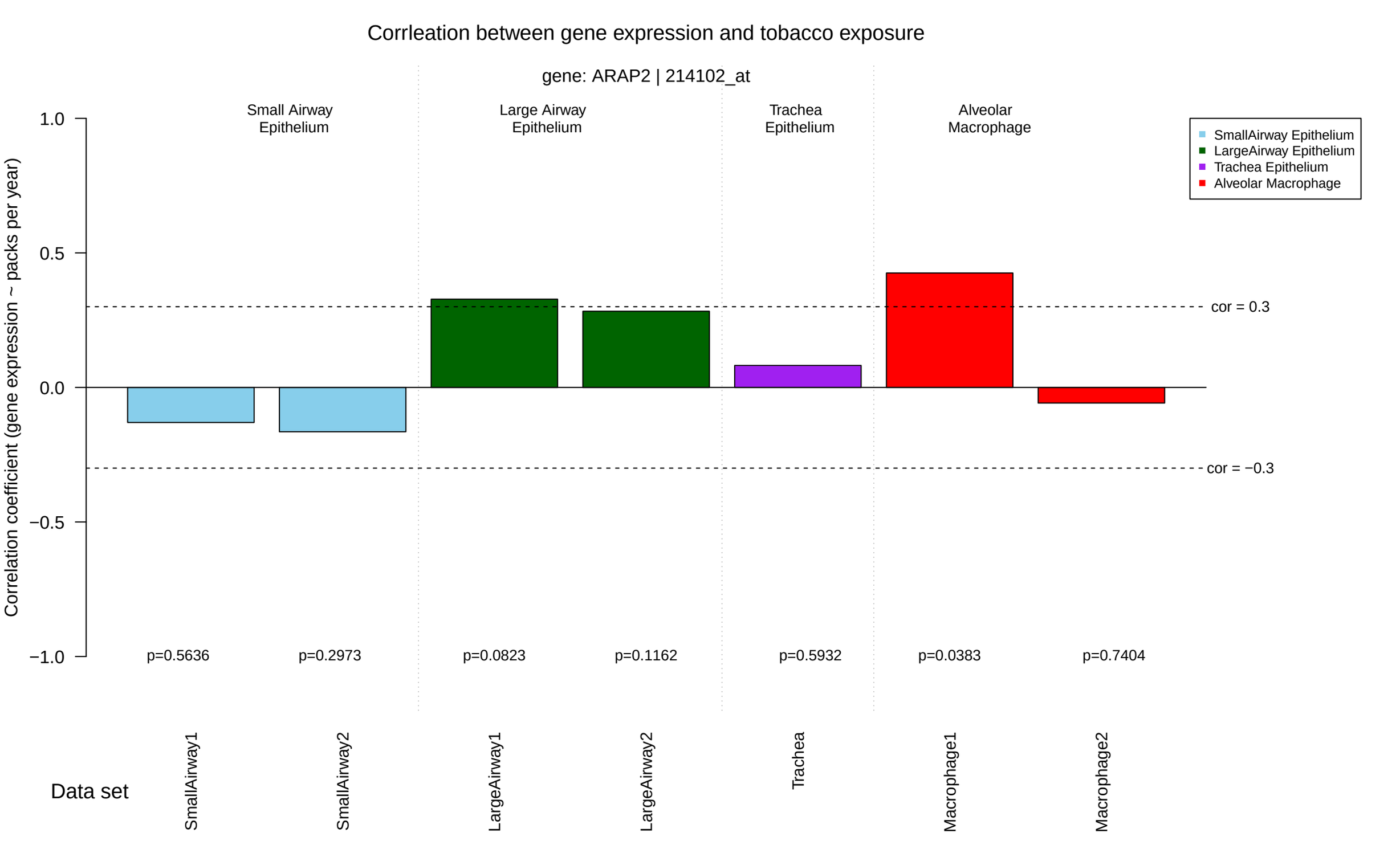

set: LargeAirway1 (p=0.2532)
ARAP2 | 214102_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9553)
ARAP2 | 214102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4029)
ARAP2 | 214102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0599)
ARAP2 | 214102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4894)
ARAP2 | 214102_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARAP2 | 214102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5636
p=0.2973
p=0.0823
p=0.1162
p=0.5932
p=0.0383
p=0.7404
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2